

Breaking up and crushing clusters: collected methods

Adapted from the lab compendium Snowflake Yeast Protocols — Ratcliff Lab / MuLTEE (V2).

Breaking up / crushing clusters

There are multiple ways to break up multicellular snowflake yeast clusters, with the best method depending on your application. The relevant steps from each protocol are collected below.

Pipetting to break up clusters before transformation

From [[Lithium acetate/PEG transformation of snowflake yeast > Cell prep]], step 5.

Add 100 μL water and pipette up and down to break up clusters until the mixture is evenly suspended (light cream color, no visible individual colonies). Later-generation clusters require more vigorous pipetting.

Mixing technique in PEG/LiOAc during transformation

From [[Lithium acetate/PEG transformation of snowflake yeast > Assembling the transformation mix]], step 2.

- Pipette up cells from bottom of solution and release at the top.
- Swirl pipette around the sides of the tube on the way back down to the bottom to mix.
- Repeat until large clusters crushed and solution uniform.

Dissociating and crushing between slides for spheroplasts

From [[Isolating individual spheroplasts for single-cell RNA extraction]], steps 4-5.

- **Step 4.** Dissociate clusters by pipetting up and down ~ 50 times using a 100 μL wide-bore pipette tip.
- **Step 5.** Gently crush clusters between two microscope slides: use the 100 μL wide-bore pipette tip to drop 10 μL of cluster pellet onto ~ 15 spots on a microscope slide, then gently place a second microscope slide on top of the first, then place a third slide on top of the second, perpendicular to it, forming a cross shape. Press lightly, then very gently remove the third and second slides.

Breaking up and single-layer crushing for cell volume

From [[Measuring cell volume]], steps 2 and 5.

- **Step 2.** Macroscopic strains: sample using 100 μL wide-bore tips, then break clusters into small pieces using 100 μL regular-bore tips to facilitate single-layer crushing later.
- **Step 5.** Crush 5 μL of stained clusters into a single layer of cells between a slide and coverslip, without shearing.

Breaking up and crushing for ploidy imaging

From [[Measuring ploidy (propidium iodide staining)]], steps 3 and 9.

- **Step 3.** Macroscopic strains: break clusters into small pieces by vigorous pipetting with 100 μ L regular-bore tips before washing.
- **Step 9.** Crush 5 μ L of stained clusters into a single cell layer between slide and coverslip for imaging.